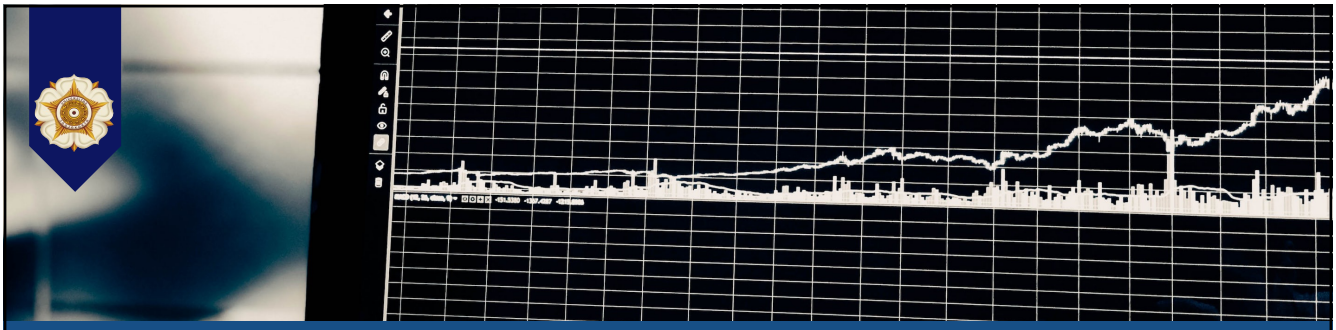


Random Sampling and Estimation (Part 2)

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Version: August 2024

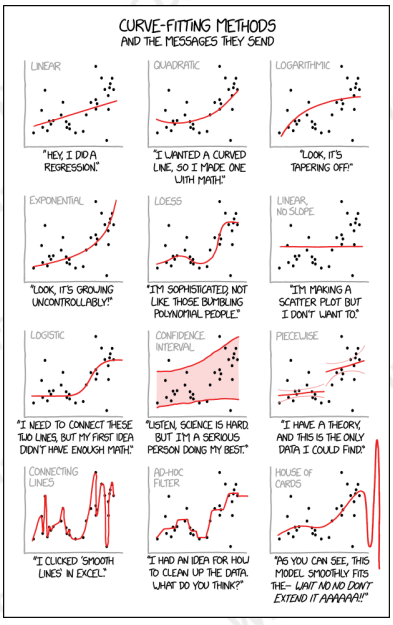
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1

Overview of today's lecture

- Random sampling
- Central limit theorem
- Estimation of population: large sample size
- Estimation of population: small sample size

(+) DataCamp assignment (Mas Adi)



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Part 3

ESTIMATION OF POPULATION: LARGE SAMPLE SIZE

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Intro to estimation

- As explained previously, populations are characterized by numerical descriptive measures called **parameters**.
- Inferences about parameter values are based on statistics computed from the information of sample.
- In this course, we will show how to estimate populations' means, variances, etc.
- To do so, we need point estimate.

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Point estimate

- A ***point estimate*** a single value computed from the observations in a sample, that is used to estimate the value of the target parameter.
- In the other words, **we need to be able to state how close our estimate is likely to be to the true value of the population.**
- For example, the sample mean $\overline{x}$ is a **point estimate** of the population mean μ
- **Solution:** using the characteristics of the sampling distribution of the statistic that was used to obtain the point estimate

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The Central Limit Theorem

- If the sample size is sufficiently large, the mean of a random sample has a sampling distribution that is approximately normal (i.e. close to normal/Gaussian distribution), regardless of the shape of the relative frequency distribution of the target population.
- As the sample size increases, the better will be the normal approximation to the sampling distribution

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Confidence interval in the Central Limit Theorem

- A ***confidence interval*** for a parameter is an interval of numbers **within which we expect the true value of the population parameter to be contained.**
- The endpoints of the interval are computed based on sample information.

$$\bar{x} \pm 1.96\sigma_{\bar{x}} = \bar{x} \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$$

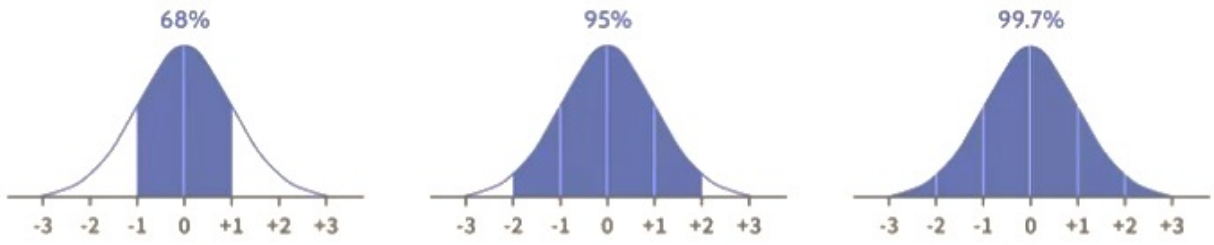
Confidence interval

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Confidence interval in the Central Limit Theorem

The construction of a confidence interval relies on the principal of the central limit theorem



68% of all values under the curve are approximately within 1 standard deviation of mean value

95% of all values under the curve are approximately within 2 standard deviation of mean value

99% of all values under the curve are approximately within 3 standard deviation of mean value

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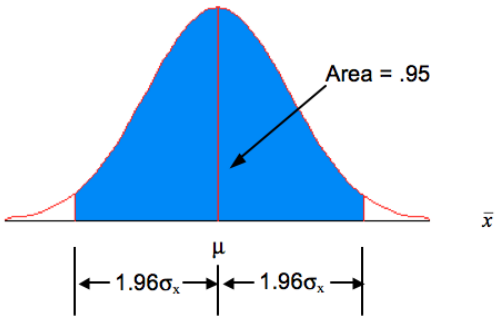
Estimation of population: Large sample case (n>=30)

- Suppose we plan to take a sample of n = 30 measurements from population of numbers of children ever born in a country and construct interval (based on central limit theorem):

$$\bar{x} \pm 1.96\sigma_{\bar{x}} = \bar{x} \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$$

Sampling distribution's standard deviation is also called by **standard error**

How can we get the coefficient 1.96? Check the next slide



95% of the area under the normal distribution lies within 1.96 standard deviations of the mean.

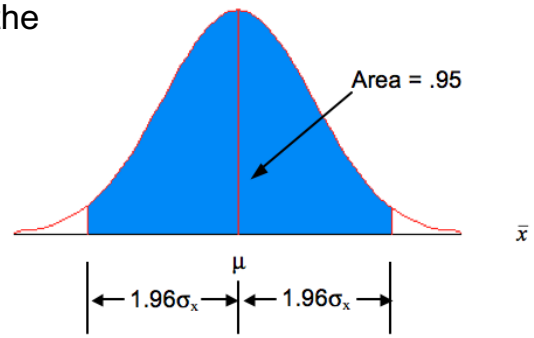
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How to compute the coefficient?

We know from central limit theorem that we want to obtain 95% of the area under the curve. Therefore:
P(… < Z < …) = 0.95
P(Z < 1.96) = 0.975
P(Z < - 1.96) = 0.025

Thus,
P(-1.96 < Z < 1.96)
= P(Z < 1.96) – P(Z<-1.96)
= 0.975 – 0.025
= 0.95



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P(Z < - 1.96) = 0.025

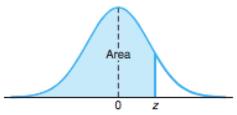


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559

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P(Z < 1.96) = 0.975

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

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Step-by-step reasoning on estimating mean

Step 1 First note that, the area beneath the sampling distribution of $\bar{x}$ between $\mu - 1.96\sigma_{\bar{x}}$ and $\mu + 1.96\sigma_{\bar{x}}$ is approximately .95. (This area colored green in Figure 7.1, is obtained from Table A.3 Appendix A) This applies that before the sample of measurements is drawn, the probability that $\bar{x}$ will fall within the interval $\mu \pm 1.96\sigma_{\bar{x}}$.

Step 2 If in fact the sample yields a value of $\bar{x}$ that falls within the interval $\mu \pm 1.96\sigma_{\bar{x}}$, then it is true that $\bar{x} \pm 1.96\sigma_{\bar{x}}$ will contain μ, as demonstrated in Figure 7.2. For particular value of $\bar{x}$ that falls within the interval $\mu \pm 1.96\sigma_{\bar{x}}$, a distance of $1.96\sigma_{\bar{x}}$ is marked off both to the left and to the right of $\bar{x}$. You can see that the value of μ must fall within $\bar{x} \pm 1.96\sigma_{\bar{x}}$.

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Solution

Step 3 Step 1 and Step 2 combined imply that, before the sample is drawn, the probability that the interval $\bar{x} \pm 1.96\sigma_{\bar{x}}$ will enclose μ is approximately .95.

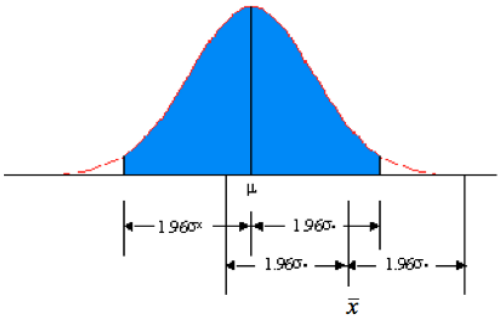


Figure 7.2 *Sample distribution of $\bar{x}$ in Example 7.2*

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Large sample

- The interval $\bar{x} \pm 1.96\sigma_{\bar{x}}$ in the previous example is called **a large-sample 95% confidence interval for the population mean.**
- The term *large-sample* refers to the sample being of a sufficiently large size that we can apply the Central Limit Theorem to determine the form of the sampling distribution $\bar{x}$.
- ***Large samples: n ≥30***
- For sufficient large sample size, the sampling distribution of the sample mean ($\bar{x}$) is approximately normal

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Confidence coefficient (1-α)

- Suppose you want to construct an interval that you believe will contain μ with some degree of confidence other than 95%
- In other words, you want to choose a confidence coefficient (1-α) other than .95.
- Alpha (α): probability of stating that observation results on the sample **do not contain the true population mean** (μ)

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Commonly used confidence coefficient

Table 7.3 *Commonly used confidence coefficient*

Confidence Coefficient		
$(1-\alpha)$	$\alpha/2$	$z_{\alpha/2}$
.90	.050	1.645
.95	.025	1.960
.98	.010	2.330
.99	.005	2.58

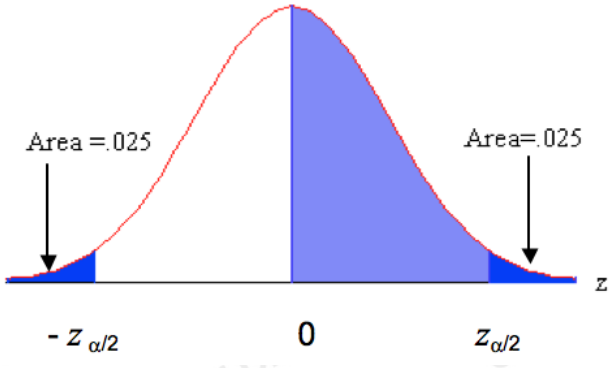
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Finding the border of confidence coefficient

Large-sample $(1-\alpha)100\%$ confidence interval for a population mean, μ

$$\bar{x} \pm z_{\alpha/2}\sigma_{\bar{x}} = \bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$



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Find

$$z_{\alpha/2} = z_{\pm 0.025} = \cdots\cdots$$

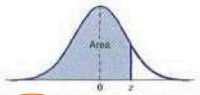


Table A.3 Areas under the Normal Curve

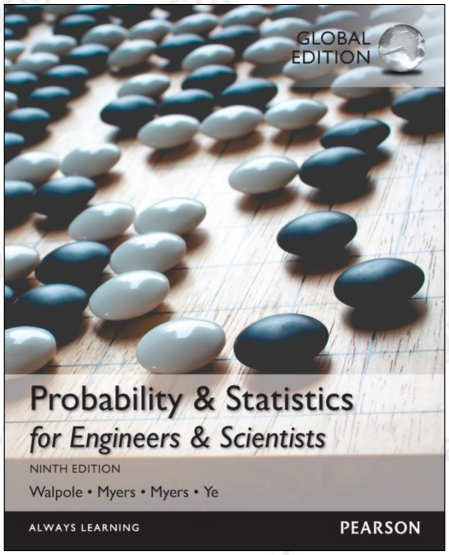
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233

$$z_{\alpha/2} = z_{\pm 0.025} = \pm 1.96$$

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We will do some examples from Walpole's book



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The Central Limit Theorem formula

(from Walpole's book p. 254)

Theorem 8.2: **Central Limit Theorem:** If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2, then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \to \infty$, is the standard normal distribution $n(z; 0, 1)$.

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Sampling distributions of the difference between two samples

(Taken from Walpole's book p. 257)

Suppose that we have two populations, the first with mean μ_1 and variance σ_1^2, and the second with mean μ_2 and variance σ_2^2. Let the statistic $\bar{X}_1$ represent the mean of a random sample of size n_1 selected from the first population, and the statistic $\bar{X}_2$ represent the mean of a random sample of size n_2 selected from the second population, independent of the sample from the first population. What can we say about the sampling distribution of the difference $\bar{X}_1 - \bar{X}_2$ for repeated samples of size n_1 and n_2? According to Theorem 8.2, the variables $\bar{X}_1$ and $\bar{X}_2$ are both approximately normally distributed with means μ_1 and μ_2 and variances σ_1^2/n_1 and σ_2^2/n_2, respectively. This approximation improves as n_1 and n_2 increase.

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Sampling distributions of the difference between two samples

Theorem 7.11: If $X_1, X_2, \ldots, X_n$ are independent random variables having normal distributions with means $\mu_1, \mu_2, \ldots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$, respectively, then the random variable

$$Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n$$

has a normal distribution with mean

$$\mu_Y = a_1\mu_1 + a_2\mu_2 + \cdots + a_n\mu_n$$

and variance

$$\sigma_Y^2 = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \cdots + a_n^2\sigma_n^2.$$

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Sampling distributions of the difference between two samples

By choosing independent samples from the two populations we ensure that the variables $\bar{X}_1$ and $\bar{X}_2$ will be independent, and then using Theorem 7.11, with $a_1 = 1$ and $a_2 = -1$, we can conclude that $\bar{X}_1 - \bar{X}_2$ is approximately normally distributed with mean

$$\mu_{\bar{X}_1 - \bar{X}_2} = \mu_{\bar{X}_1} - \mu_{\bar{X}_2} = \mu_1 - \mu_2$$

and variance

$$\sigma^2_{\bar{X}_1 - \bar{X}_2} = \sigma^2_{\bar{X}_1} + \sigma^2_{\bar{X}_2} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

The Central Limit Theorem can be easily extended to the two-sample, two-population case.

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Sampling distributions of the difference between two samples

Theorem 8.3: If independent samples of size n_1 and n_2 are drawn at random from two populations, discrete or continuous, with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2, respectively, then the sampling distribution of the differences of means, $\bar{X}_1 - \bar{X}_2$, is approximately normally distributed with mean and variance given by

$$\mu_{\bar{X}_1-\bar{X}_2} = \mu_1 - \mu_2 \text{ and } \sigma^2_{\bar{X}_1-\bar{X}_2} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Hence,

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{(\sigma_1^2/n_1) + (\sigma_2^2/n_2)}}$$

is approximately a standard normal variable.

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Example #1 (Walpole's book)

(from Walpole's book p. 258-259)

Case Study 8.2: **Paint Drying Time**: Two independent experiments are run in which two different types of paint are compared. Eighteen specimens are painted using type A, and the drying time, in hours, is recorded for each. The same is done with type B. The population standard deviations are both known to be 1.0.

Assuming that the mean drying time is equal for the two types of paint, find $P(\bar{X}_A - \bar{X}_B > 1.0)$, where $\bar{X}_A$ and $\bar{X}_B$ are average drying times for samples of size $n_A = n_B = 18$.

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Example #1 (Walpole's book)

(from Walpole's book p. 258-259)

***Solution*:** From the sampling distribution of $\bar{X}_A - \bar{X}_B$, we know that the distribution is approximately normal with mean

$$\mu_{\bar{X}_A - \bar{X}_B} = \mu_A - \mu_B = 0$$

and variance

$$\sigma^2_{\bar{X}_A - \bar{X}_B} = \frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B} = \frac{1}{18} + \frac{1}{18} = \frac{1}{9}.$$

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Example #1 (Walpole's book)

(from Walpole's book p. 258-259)

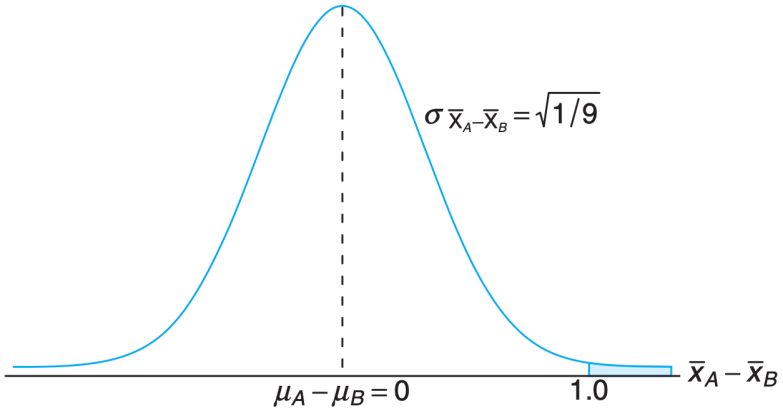


Figure 8.5: Area for Case Study 8.2.

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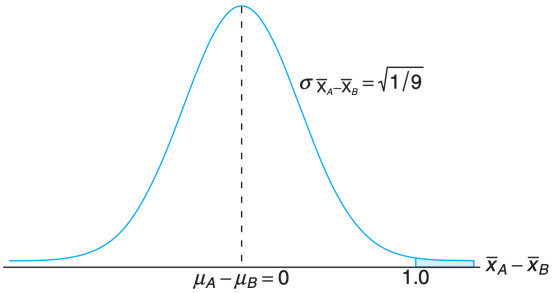


Figure 8.5: Area for Case Study 8.2.

The desired probability is given by the shaded region in Figure 8.5. Corresponding to the value $\bar{X}_A - \bar{X}_B = 1.0$, we have

$$z = \frac{1-(\mu_A-\mu_B)}{\sqrt{1/9}} = \frac{1-0}{\sqrt{1/9}} = 3.0;$$

so

$$P(Z > 3.0) = 1 - P(Z < 3.0) = 1 - 0.9987 = 0.0013.$$

Table A.3 (continued) Areas under the Normal Curve

3.0	0.9987
3.1	0.9990
3.2	0.9993
3.3	0.9995
3.4	0.9997

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Example #2 (Walpole's book)

(from Walpole's book p. 260)

Example 8.6: The television picture tubes of manufacturer A have a mean lifetime of 6.5 years and a standard deviation of 0.9 year, while those of manufacturer B have a mean lifetime of 6.0 years and a standard deviation of 0.8 year. What is the probability that a random sample of 36 tubes from manufacturer A will have a mean lifetime that is at least 1 year more than the mean lifetime of a sample of 49 tubes from manufacturer B?

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Example #2 (Walpole's book)

(from Walpole's book p. 260)

***Solution*:** We are given the following information:

Population 1	Population 2
$\mu_1 = 6.5$	$\mu_2 = 6.0$
$\sigma_1 = 0.9$	$\sigma_2 = 0.8$
$n_1 = 36$	$n_2 = 49$

If we use Theorem 8.3, the sampling distribution of $\bar{X}_1 - \bar{X}_2$ will be approximately normal and will have a mean and standard deviation

$$\mu_{\bar{X}_1-\bar{X}_2} = 6.5 - 6.0 = 0.5 \quad \text{and} \quad \sigma_{\bar{X}_1-\bar{X}_2} = \sqrt{\frac{0.81}{36} + \frac{0.64}{49}} = 0.189.$$

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Example #2 (Walpole's book)

(from Walpole's book p. 260)

The probability that the mean lifetime for 36 tubes from manufacturer A will be at least 1 year longer than the mean lifetime for 49 tubes from manufacturer B is given by the area of the shaded region in Figure 8.6. Corresponding to the value $\bar{x}_1 - \bar{x}_2 = 1.0$, we find that

$$z = \frac{1.0 - 0.5}{0.189} = 2.65,$$

and hence

$$P(\bar{X}_1 - \bar{X}_2 \geq 1.0) = P(Z > 2.65) = 1 - P(Z < 2.65)$$
$$= 1 - 0.9960 = 0.0040.$$

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974

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Example #2 (Walpole's book)

(from Walpole's book p. 260)

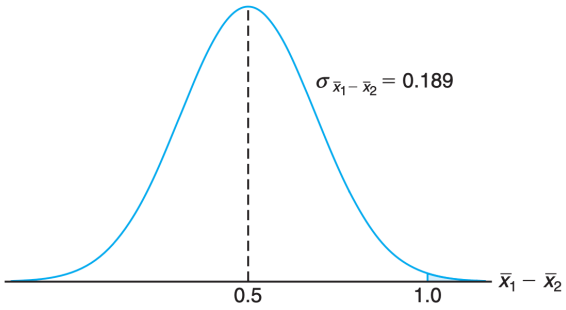


Figure 8.6: Area for Example 8.6.

$$P(\bar{X}_1 - \bar{X}_2 \geq 1.0) = P(Z > 2.65) = 1 - P(Z < 2.65)$$
$$= 1 - 0.9960 = 0.0040.$$

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Part 4

ESTIMATION OF POPULATION: SMALL SAMPLE SIZE

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Problems

- Limited time and cost may cause the amount of sample is not large enough
- How can we solve this problem? Does Central Limit Theorem can be used for small sample?

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Small sample

- Since the Central Limit Theorem **applies only to large samples**, we are not able to assume that the sampling distribution of $\overline{x}$ is approximately normal.
- For small samples, the sampling distribution of $\overline{x}$ depends on the particular form of the relative frequency distribution of the population being sampled.
- The sample standard deviation *s* **may not be a satisfactory approximation** to the population standard deviation σ if the sample size is small.

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We can use small sample estimation if….

- The amount of samples (n) < 30
  OR
- Variance of the population is unknown

With assumption:
The population from which the sample is selected has an approximate normal distribution. So we have to know the distribution of population before we do small sampling estimation.

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Small-sample confidence interval for μ

$$\bar{x} \pm t_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$$

t based on (n - 1) degrees of freedom

s = standard deviation of sample
n = amount of sample (less than 30)

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t-distribution

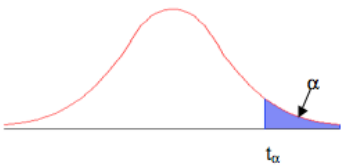
- The *t*-distribution is very much like the z-distribution. In particular, both are symmetric, bell-shaped, and have a mean of 0.
- However, the distribution of *t* depends on a quantity called its **degrees of freedom** (*df*), which is equal to (*n* - *1*) when estimating a population mean based on a small sample of size *n*.

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Table 7.4

Some values for Student's t-distribution



Degrees of freedom	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$	$t_{.0005}$
1	3.078	6.314	12.706	31.821	63.657	318.31	636.62
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.102	3.852	4.221
14	1.345	1.760	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073

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Example

Example Using Table 7.4 to determine the t-value that would be used in constructing a 95% confidence interval for μ based on a sample of size n = 14.

SOLUTION : CHECK THE t-TABLE!

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Example

Example Using Table 7.4 to determine the t-value that would be used in constructing a 95% confidence interval for μ based on a sample of size n = 14.

Solution For confidence coefficient of .95, we have

$$1-\alpha = .95$$
$$\alpha = .05$$
$$\alpha/2 = .025$$

We require the value of $t_{.025}$ for a t-distribution based on $(n - 1) = (14 - 1) = 13$ degrees of freedom. In Table 7.4, at intersection of the column labeled $t_{.025}$ and the row corresponding to df = 13, we find the entry 2.160 (see Figure 7.6). Hence, a 95% confidence interval for μ, based on a sample of size n = 13 observations, would be given by

$$\bar{x} \pm 2.160\left(\frac{s}{\sqrt{14}}\right)$$

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Table 7.4

Some values for Student's t-distribution



Degrees of freedom	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$	$t_{.0005}$
1	3.078	6.314	12.706	31.821	63.657	318.31	636.62
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.102	3.852	4.221
14	1.345	1.760	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073

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Relationship between *t* and *z* distributions

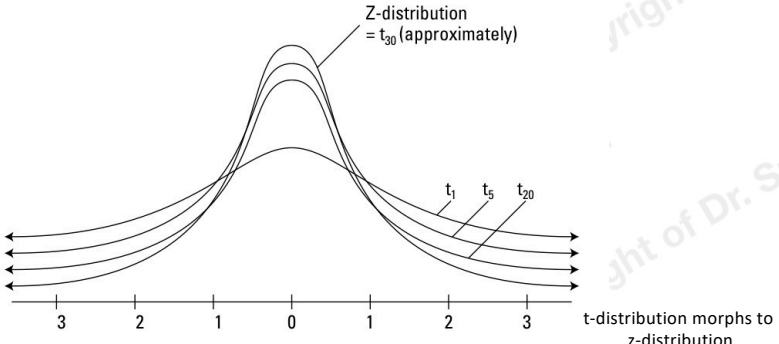


Figure 9-1: *t*-distributions for different sample sizes

t-Table

df/p	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0005
29	0.255684	0.683044	1.311434	1.699127	2.04523	2.46202	2.75639	3.6594
30	0.255605	0.682756	1.310415	1.697261	2.04227	2.45726	2.75000	3.6460
∞	0.253347	0.674490	1.281552	1.644854	1.95996	2.32635	2.57583	3.2905

z-Table

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233

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Summary

- Confidence interval is used to state how close our estimate is likely to be to the true value of the population.
- The statistician denotes that n = 30 belongs to large data sample → use standardized normal distribution (z-table). Assumption: we know the variance of population.
- If sample size is less than 30 or if we don't know variance of the population → use t distribution.
- As the sample size increases, the t-distribution becomes more like the z distribution

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End of File

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